

TEMPEST: Soil Gas Well

June 2025 Samples

2026-09-24

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##Run Information

```
cat("Run Information: Brenden Blakley") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
#set the run date & user name
```

```
sample_year <- "2025"
```

```
sample_month <- "June"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month (.txt file(s))
```

```
GHG_files <- c("Raw Data/20260827_TMP2025_June.txt", "Raw Data/20260827_TMP2025_June_Sept.txt" )
```

```
#Metadata file path
```

```
Metadata_path = "Metadata/TEMPEST_SGAS_202506_ALL_WORKING.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- "GC_QAQC_Log.csv"
```

```
check_log_path <- "GC_Check_QAQC_Log.csv"
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/TMP2025_June.csv"
```

```
##Standard Type ("Actual" or "Half")
```

```
#LTREB, SMARTX, and GENX PW Samples use "Half", others should use "Actual"
```

```
Standard_Type <- "Actual"
```

```
#Did Some Samples require manual area retrieval?
```

```
##List Samples and their areas for each analyte ("CH4", "CO2", or "N2O")
```

```
Manual_Areas <- data.frame(
```

```
  Sample = c("NA"),
```

```
  Area = c("NA"),
```

```
  Analyte = c("NA")
```

```
)
```

```
#Gas in Water Calculations
```

```
#Are All Samples Gas in Water? ("YES" or "NO")
```

```
GIW <- "NO"
```

```
#Are some samples Gas in Water?
```

```
##List Sample IDs
```

```
GIW_Samples <- c()
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- " Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected"

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)
```

```
## Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected
```

```
##Set Up
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

```
##Read In Data Files
```

```
##Get Location of Each Sample in Main Data Frame
```

```
##Turn Each Sample into it's own Data Frame
```

```
##Get Only Area Tables and Retrieve Info
```

```
##QAQC Peak Count and Retention Time
```

```
## [1] "If there are too many peaks there it's possible there is a problem with the sample or run and t
```

```
## [1] "FID: LabAir; 3"          "FID: Blank1; 0"
## [3] "FID: Blank2; 0"          "FID: Oppm; 0"
## [5] "FID: Std_10K_3mL; 1"     "FID: Std_10K_6mL; 1"
## [7] "FID: Std_10K_12mL; 1"    "FID: Std_10K_24mL; 1"
## [9] "FID: Blank3; 0"          "FID: Blank4; 0"
## [11] "FID: Blank5; 0"          "FID: 2025_June_12; 3"
## [13] "FID: 2025_June_14; 3"    "FID: Blank6; 0"
## [15] "FID: 2025_June_21; 3"    "FID: 2025_June_23; 3"
## [17] "FID: 2025_June_25; 3"    "FID: Blank7; 0"
## [19] "FID: 2025_June_32; 3"    "FID: 2025_June_33; 3"
## [21] "FID: 2025_June_38; 3"    "FID: Oppm; 1"
## [23] "FID: Std_10K_3mL; 1"     "FID: Std_10K_6mL; 1"
## [25] "FID: Std_10K_12mL; 1"    "FID: Std_10K_24mL; 1"
## [27] "FID: Blank3; 0"          "FID: 2025_June_42; 3"
## [29] "FID: 2025_June_59; 3"    "FID: Blank6; 1"
## [31] "FID: Blank7; 1"          "FID: 2025_September_12; 3"
## [33] "FID: Blank8; 1"          "FID: Blank10; 1"
## [35] "FID: Blank12; 1"
```

```
## [1] "ECD: Blank1; 0"          "ECD: Oppm; 0"          "ECD: Std_10K_3mL; 0"
## [4] "ECD: Std_10K_6mL; 0"     "ECD: Std_10K_12mL; 0"  "ECD: Std_10K_24mL; 0"
## [7] "ECD: Blank3; 0"          "ECD: Blank4; 0"          "ECD: Blank5; 0"
## [10] "ECD: Blank6; 0"          "ECD: Blank7; 0"          "ECD: Std_10K_3mL; 0"
## [13] "ECD: Std_10K_6mL; 0"     "ECD: Std_10K_12mL; 0"  "ECD: Std_10K_24mL; 0"
## [16] "ECD: Blank3; 0"          "ECD: Blank6; 0"          "ECD: Blank7; 0"
## [19] "ECD: Blank8; 0"          "ECD: Blank10; 0"        "ECD: Blank12; 0"
```

```
## [1] "If RTime is wrong check that correct number of peaks were found, if not a peak may have been mi
```

```
## [1] "CH4: "
```

```
## [1] "CO2:  "

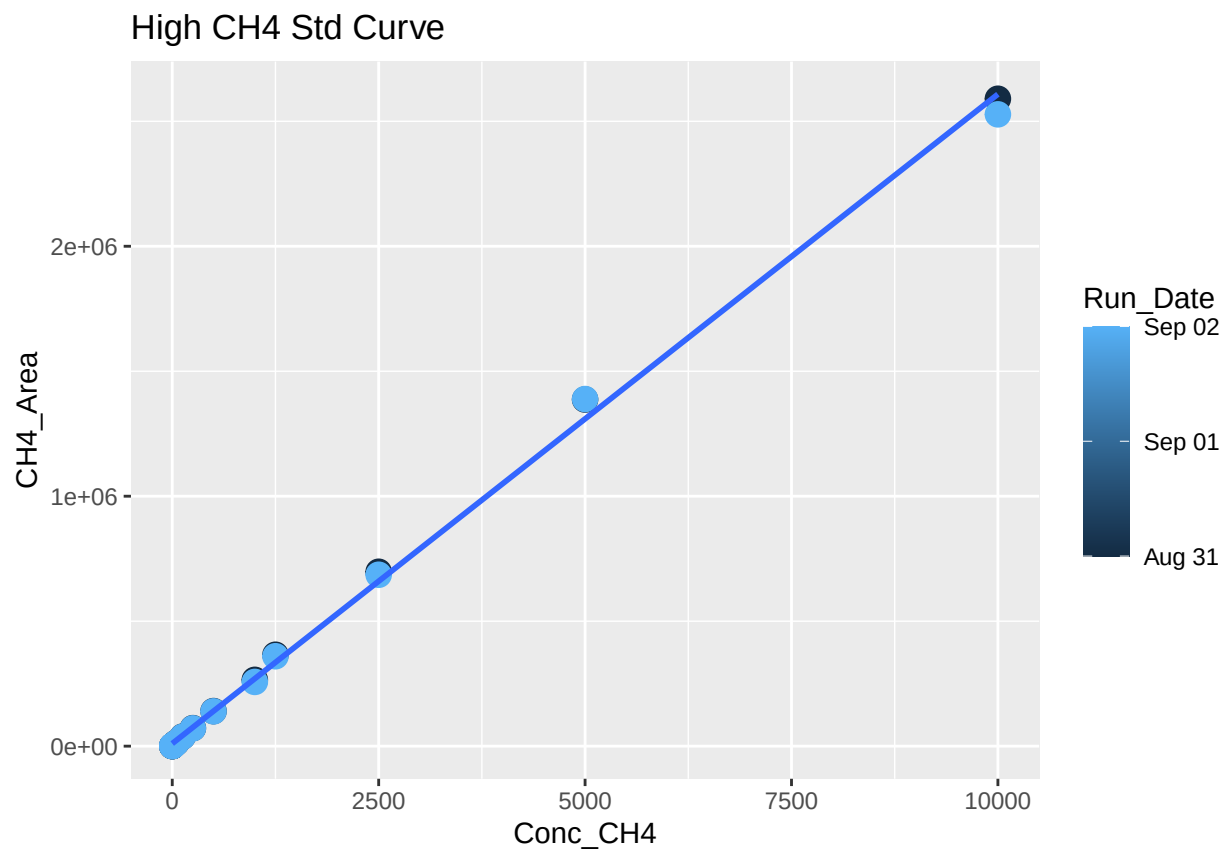
## [1] "N2O:  Blank2; 2.748"

##Format Data
##Add Standard Concentrations
##Finalize Data
```

1 Assess standard curves CH4

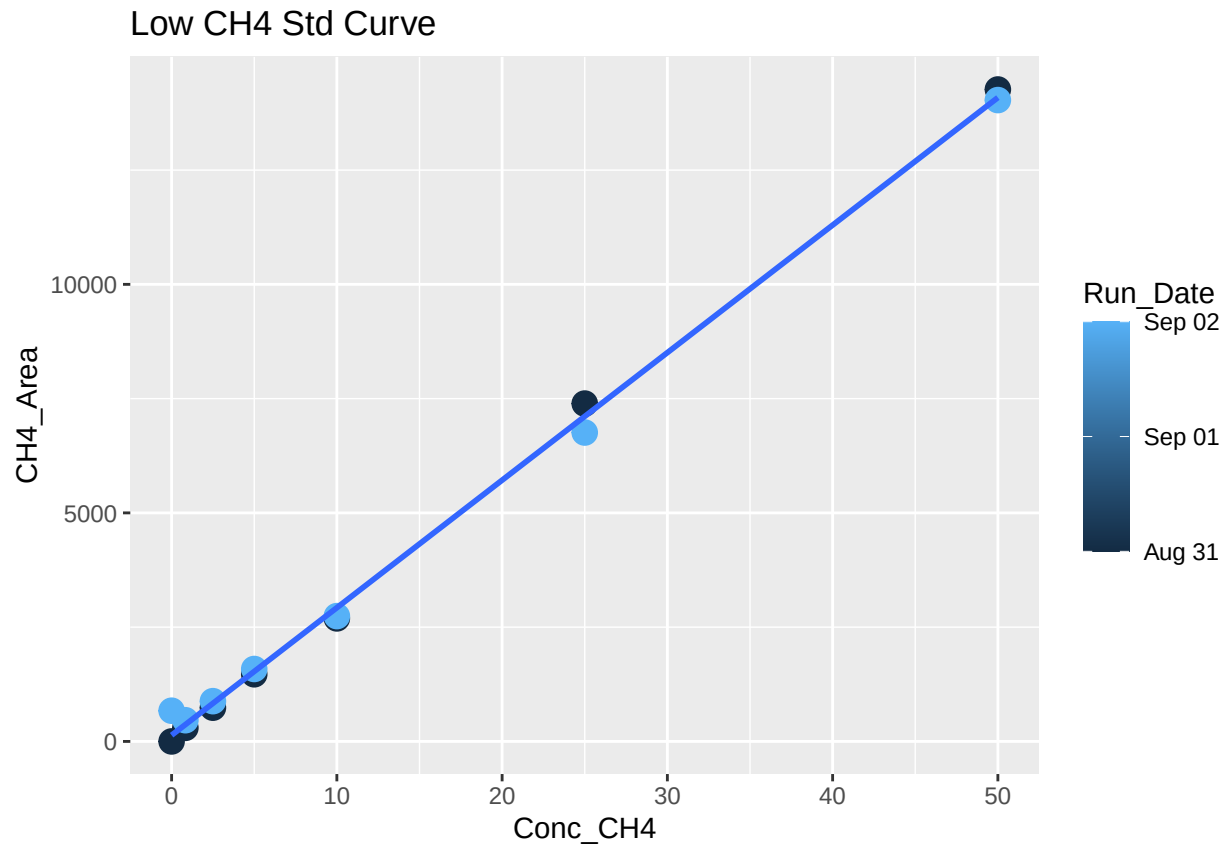
```
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```

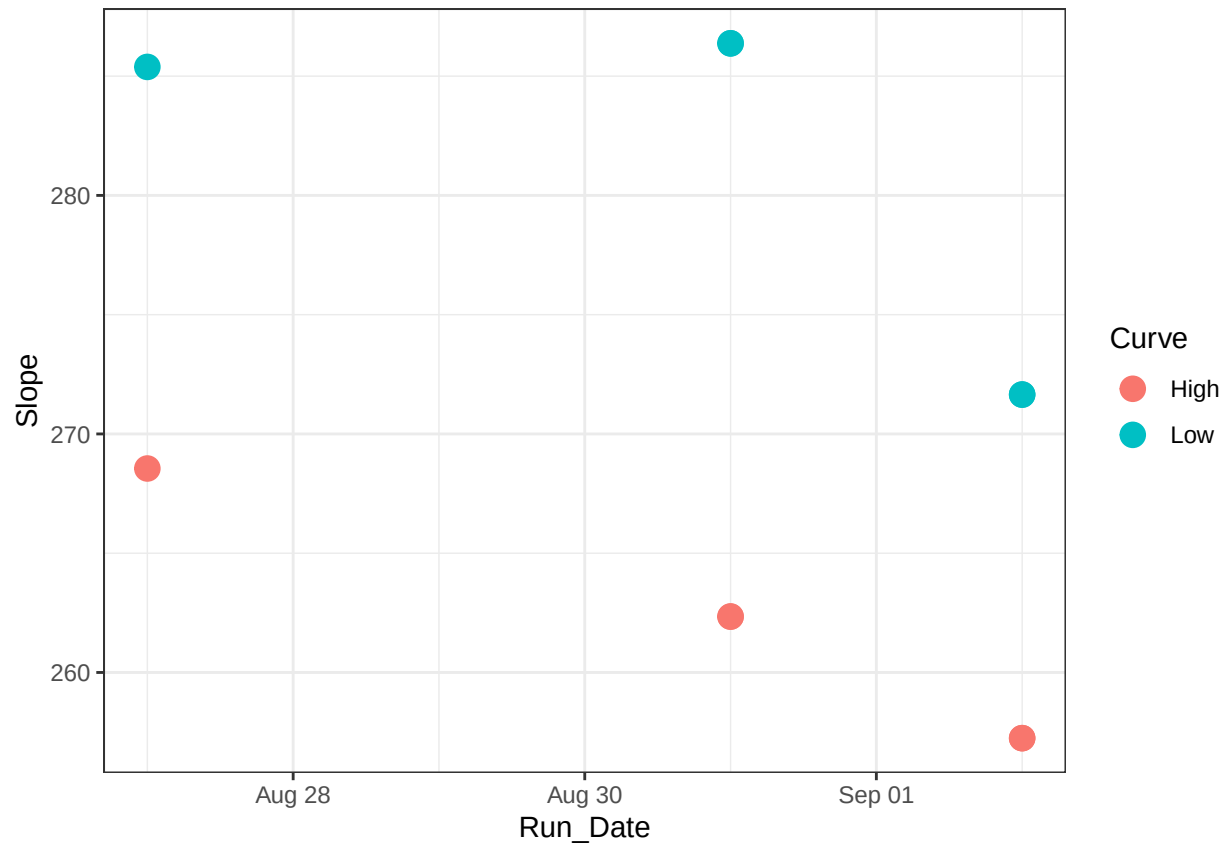


```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4



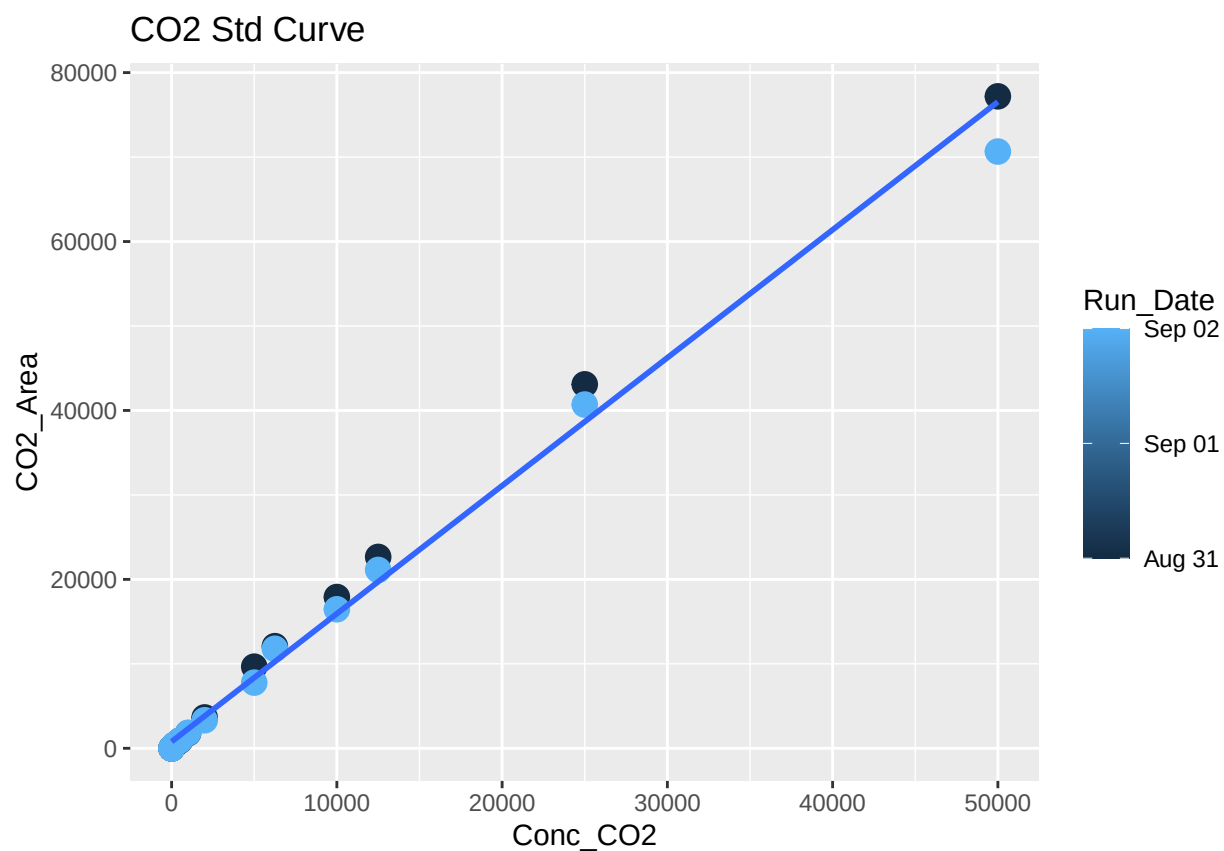
```
## [1] "High_R2 is Good-2026-08-31" "High_R2 is Good-2026-09-02"
```

```
## [1] "Low_R2 is Good-2026-08-31" "Low_R2 is Good-2026-09-02"
```

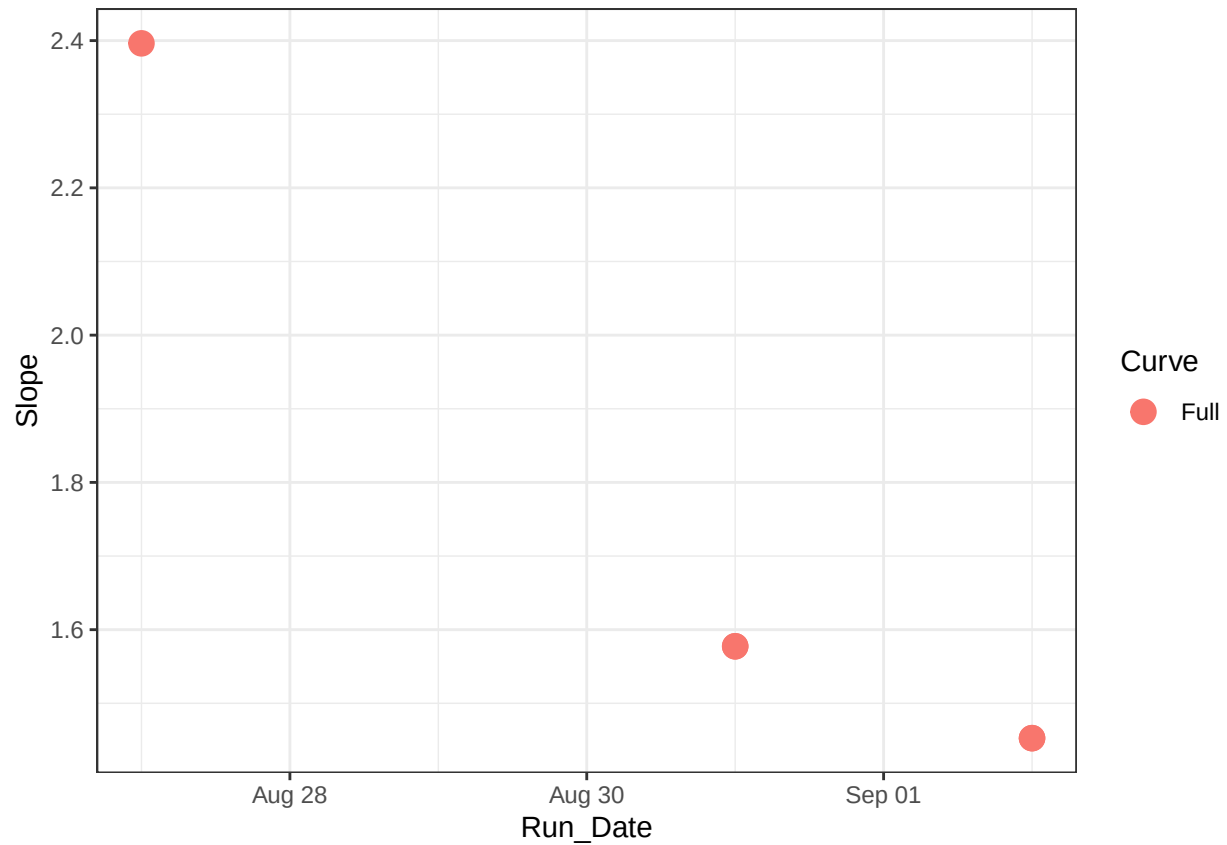
2 Assess standard curves CO2

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

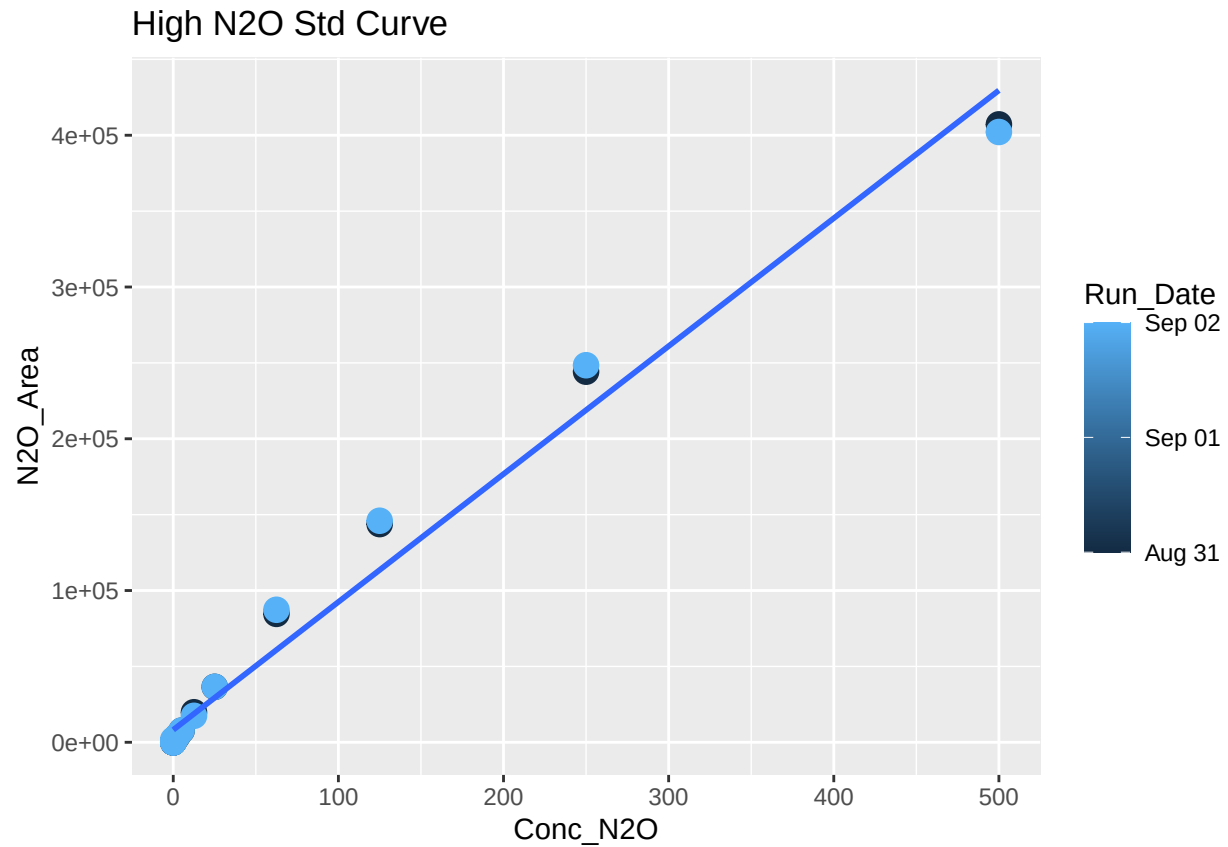


```
## [1] "C02_R2 is Good-2026-08-31" "C02_R2 is Good-2026-09-02"
```

3 Assess standard curves N2O

```
## 'geom_smooth()' using formula = 'y ~ x'
```

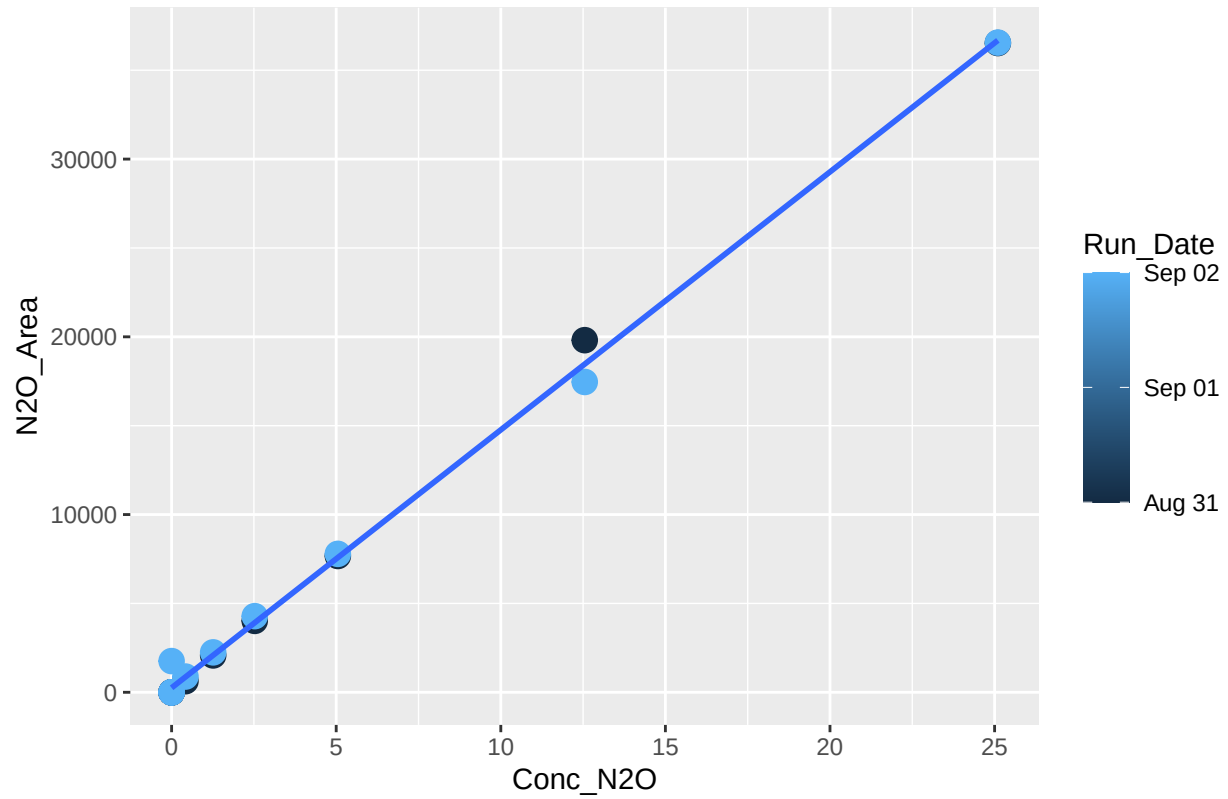
```
## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a 'group' aesthetic or to convert a numerical
## variable into a factor?
```



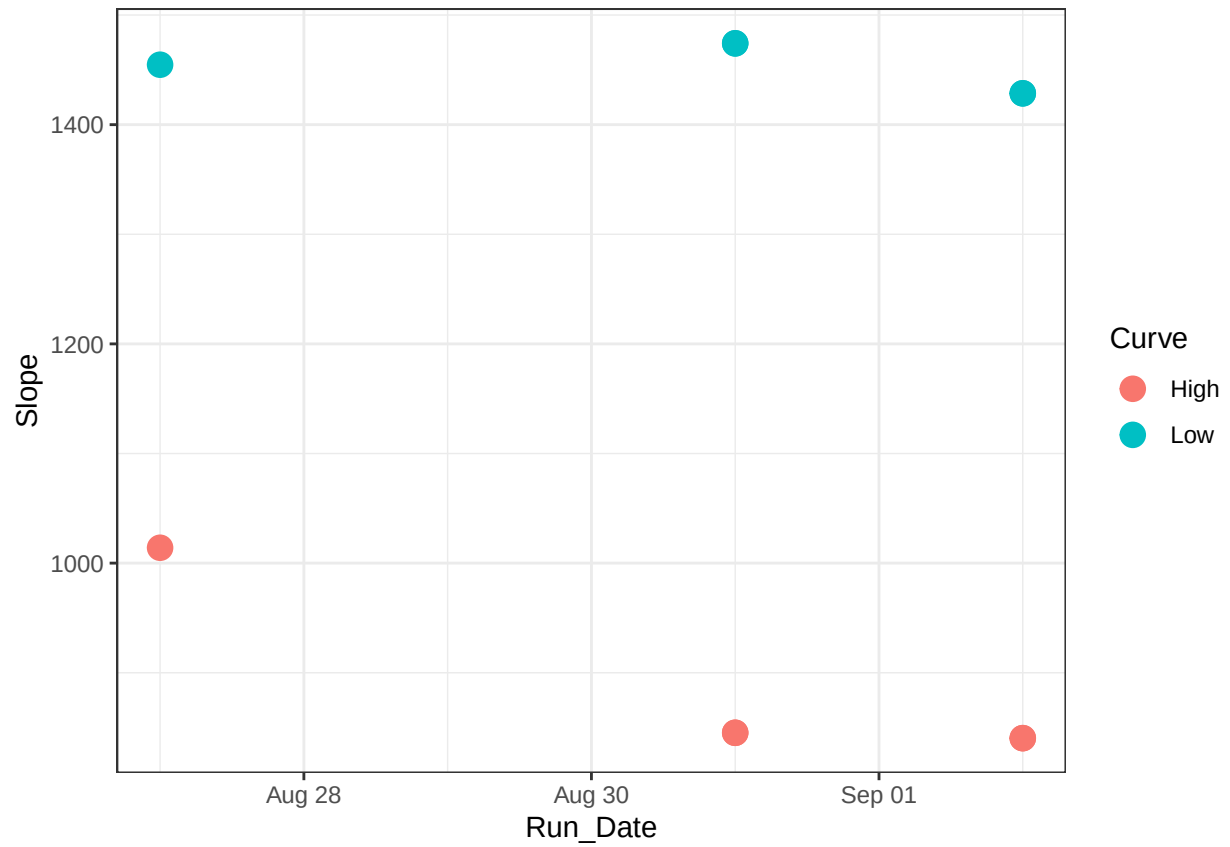
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: The following aesthetics were dropped during statistical transformation:  
## colour.  
## i This can happen when ggplot fails to infer the correct grouping structure in  
## the data.  
## i Did you forget to specify a 'group' aesthetic or to convert a numerical  
## variable into a factor?
```

Low N2O Std Curve



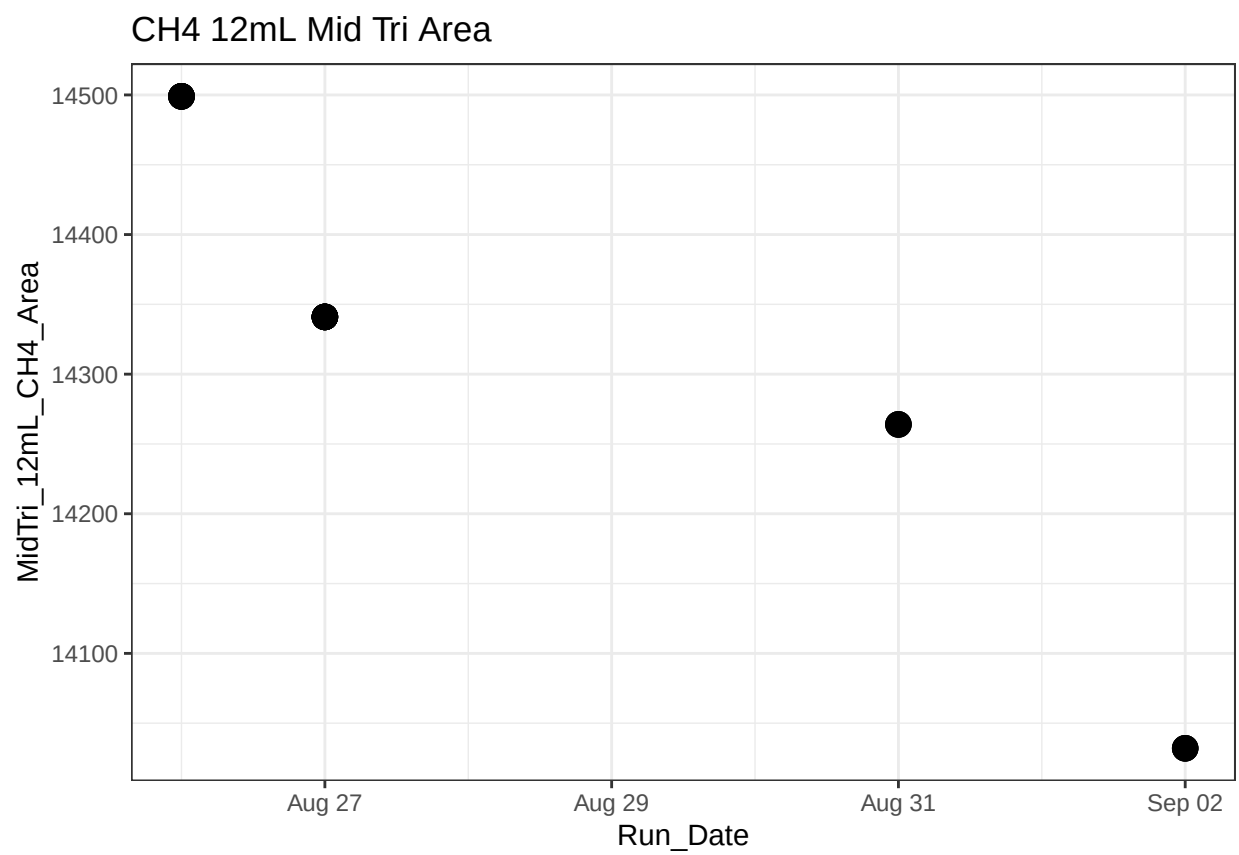
##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

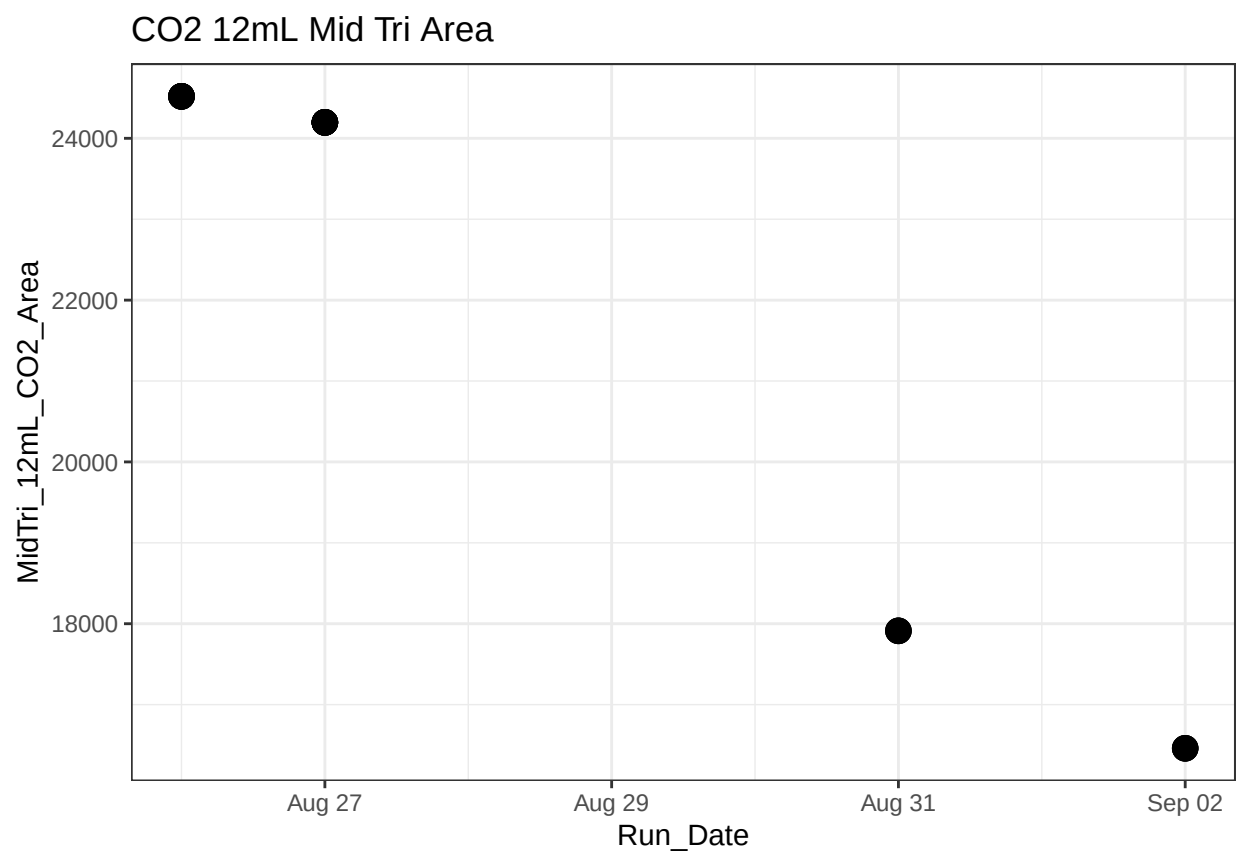


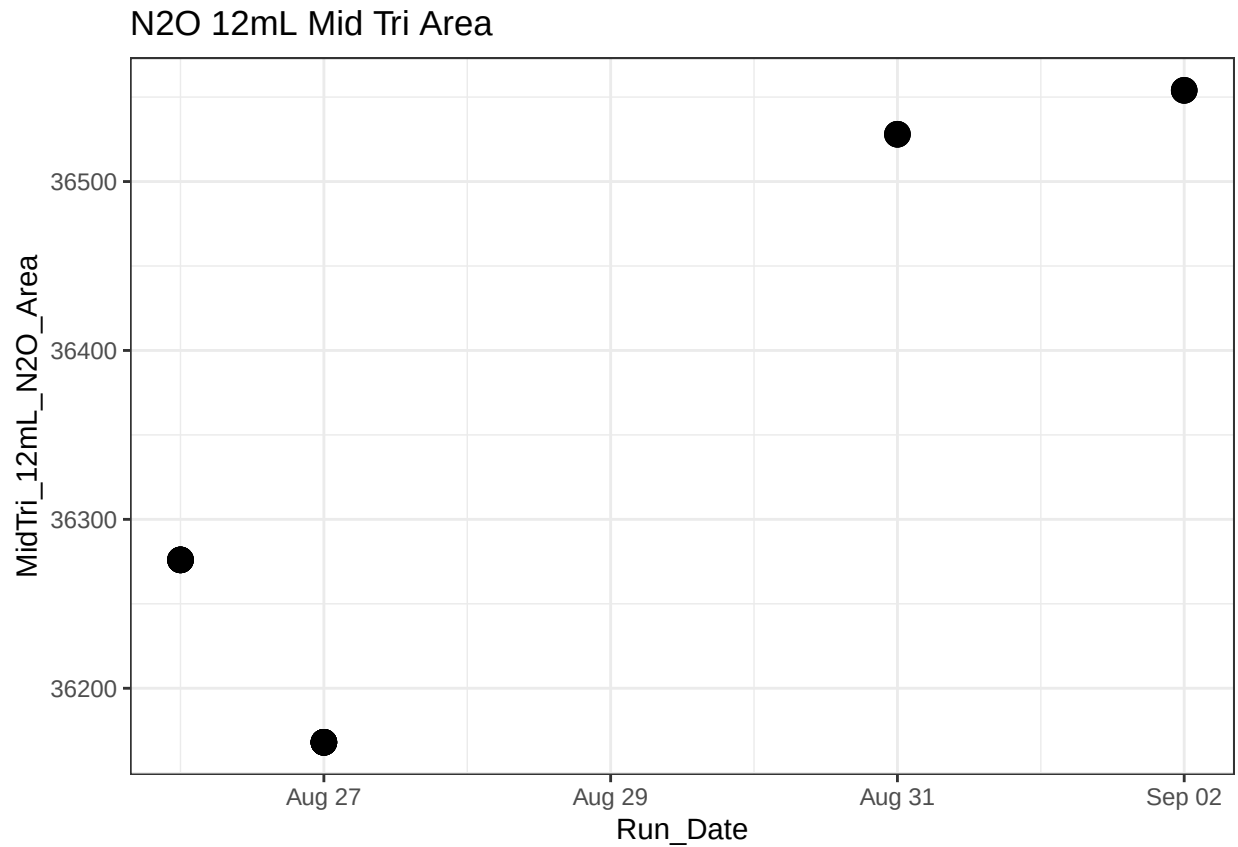
```
## [1] "High_R2 is Bad-2026-08-31" "High_R2 is Bad-2026-09-02"
```

```
## [1] "Low_R2 is Good-2026-08-31" "Low_R2 is Good-2026-09-02"
```

```
#Plot Check Areas over Time
```







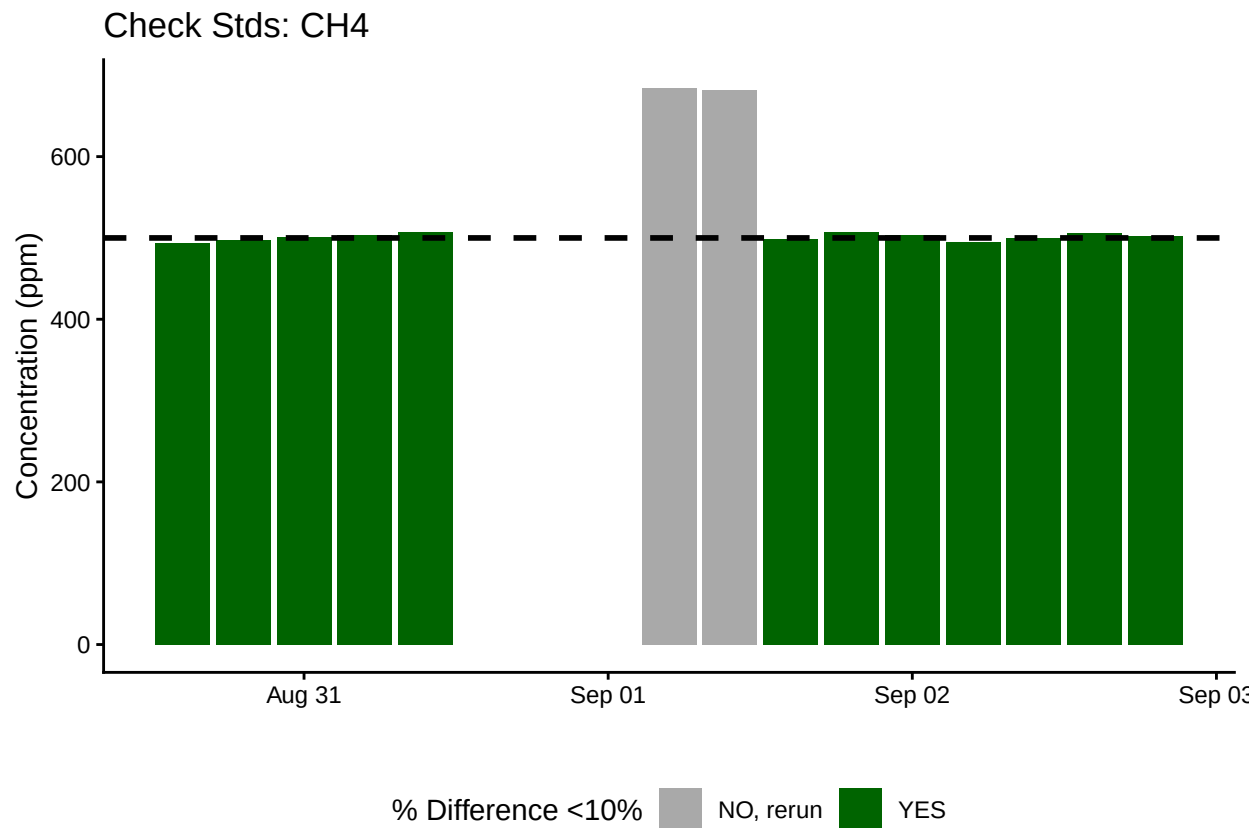
3.1 Now calculate the CH₄ in ppm

3.2 Now calculate the CO₂ in ppm

3.3 Now calculate the N₂O in ppm

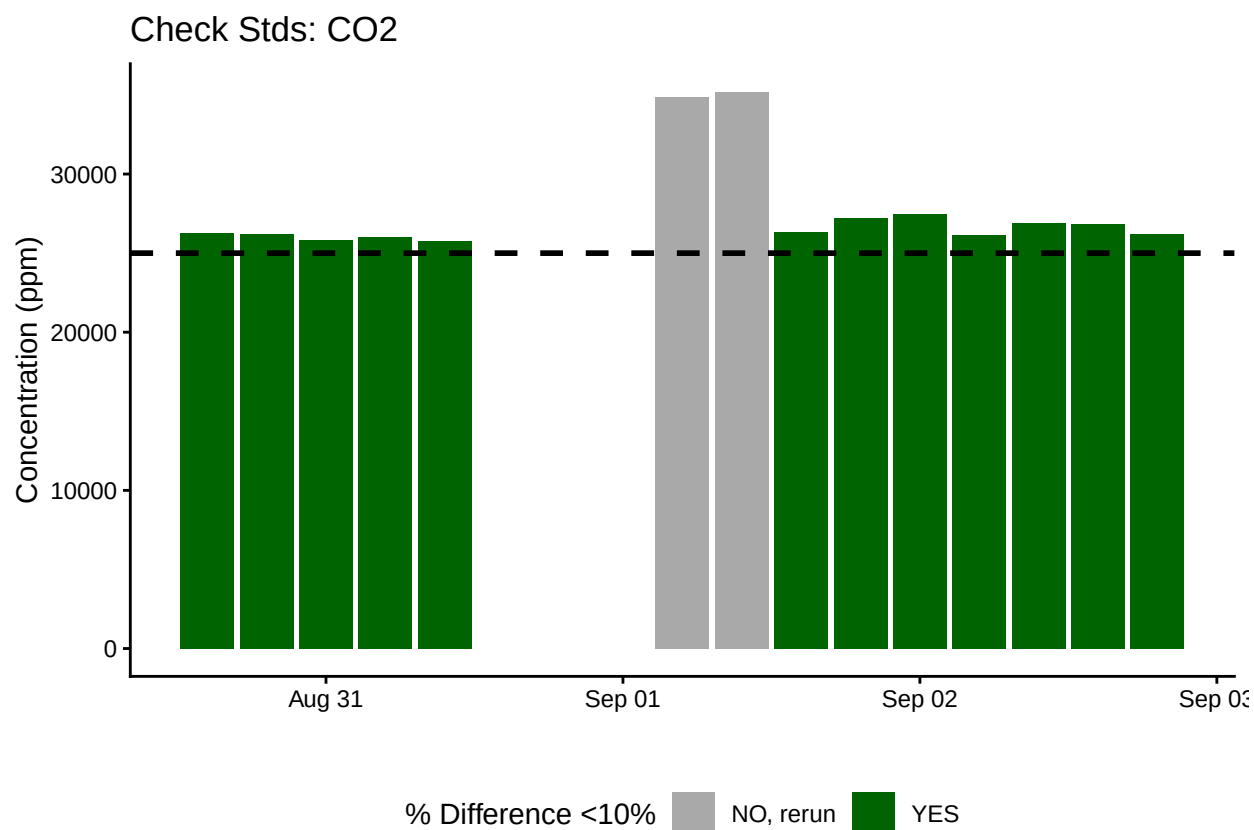
4 Aanlayze the CH₄ Check Standards

>60% of CH₄ Check Standards are within range of expected concentration - PROCEED



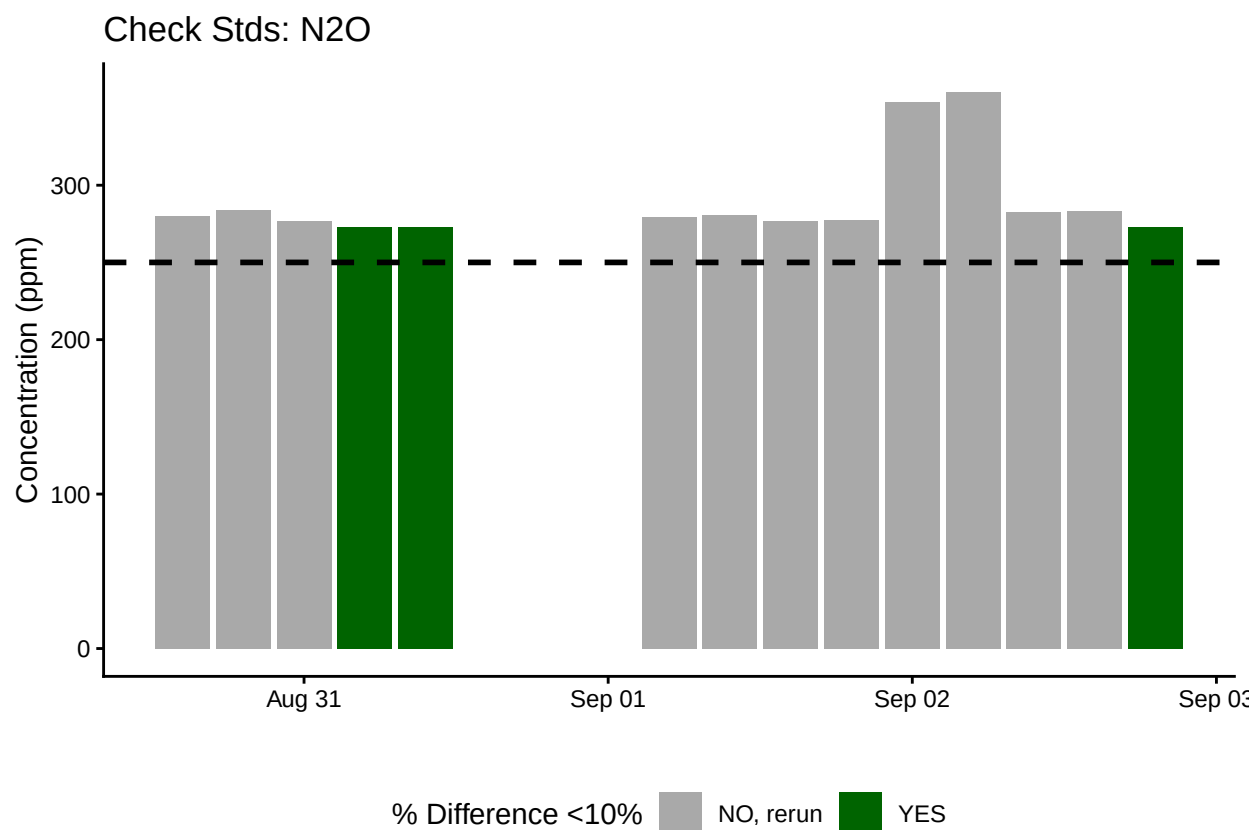
5 Aanlayze the CO2 Check Standards

>60% of CO2 Check Standards are within range of expected concentration - PROCEED



6 Aanlayze the N2O Check Standards

<60% of N2O Check Standards are within range of expected concentration - REASSESS



7 Analyze the CH₄ Blanks

Assess CH₄ Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

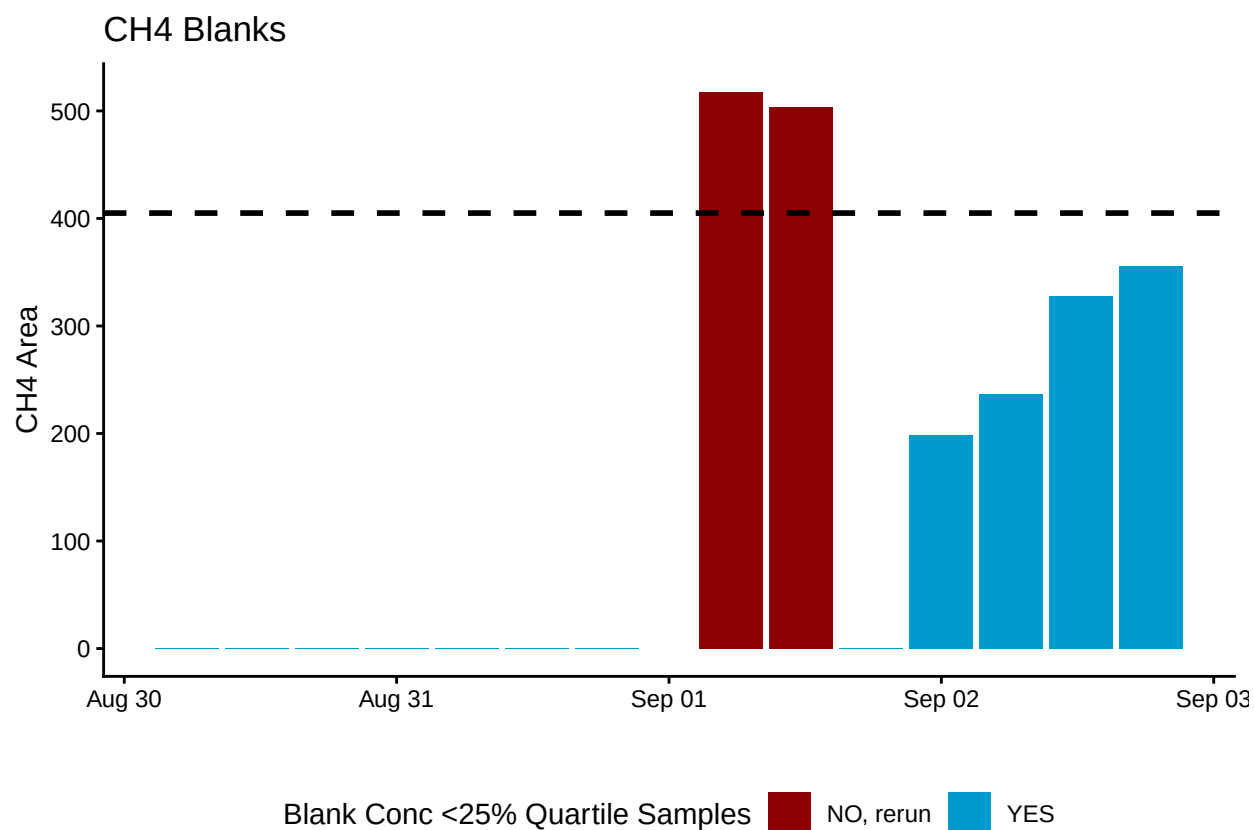


Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	153

8 Analyze the CO2 Blanks

```
## Assess CO2 Blanks
```

```
## >60% of CO2 Blanks are below the lower 25% quartile of samples - PROCEED
```

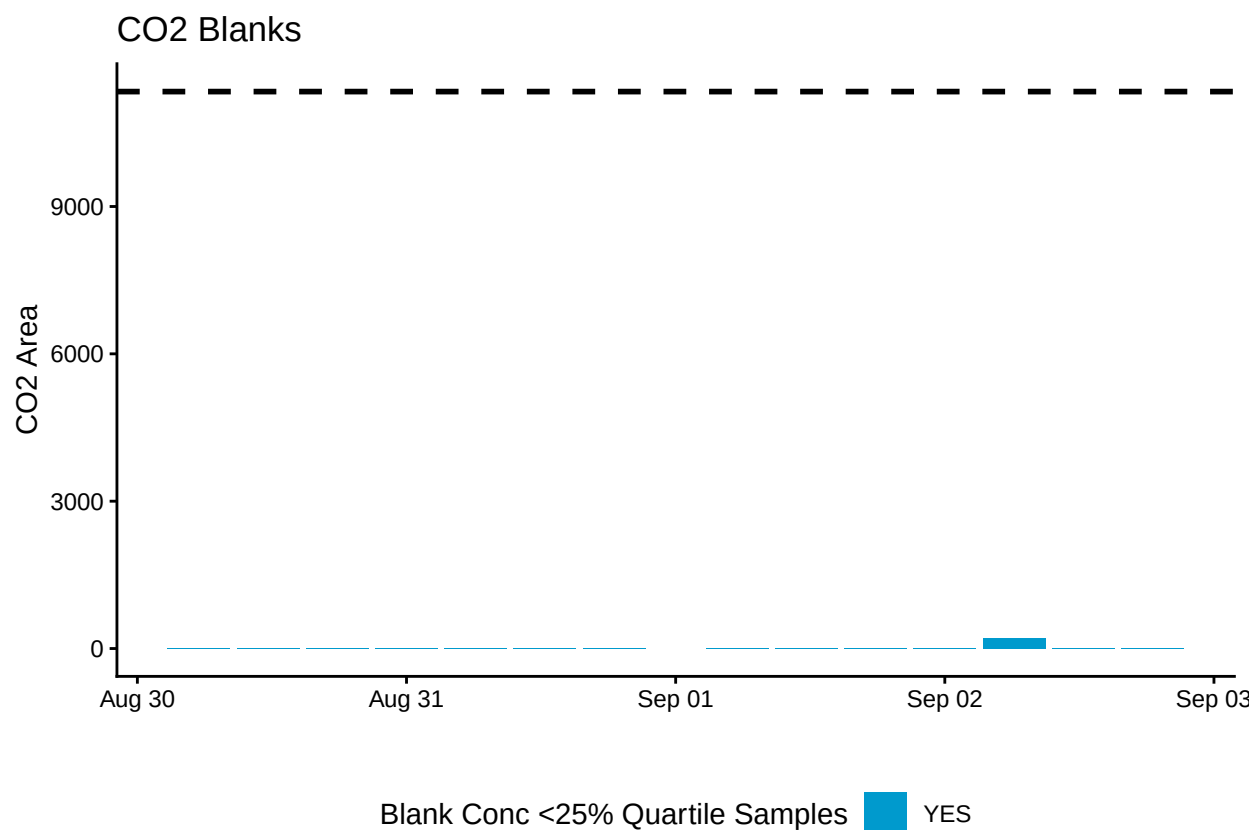


Table 2: Mean Area of Blanks

Test	Blank_Mean_Area
CO2	15.2143

9 Analyze the N2O Blanks

Assess N2O Blanks

>60% of N2O Blanks are below the lower 25% quartile of samples - PROCEED

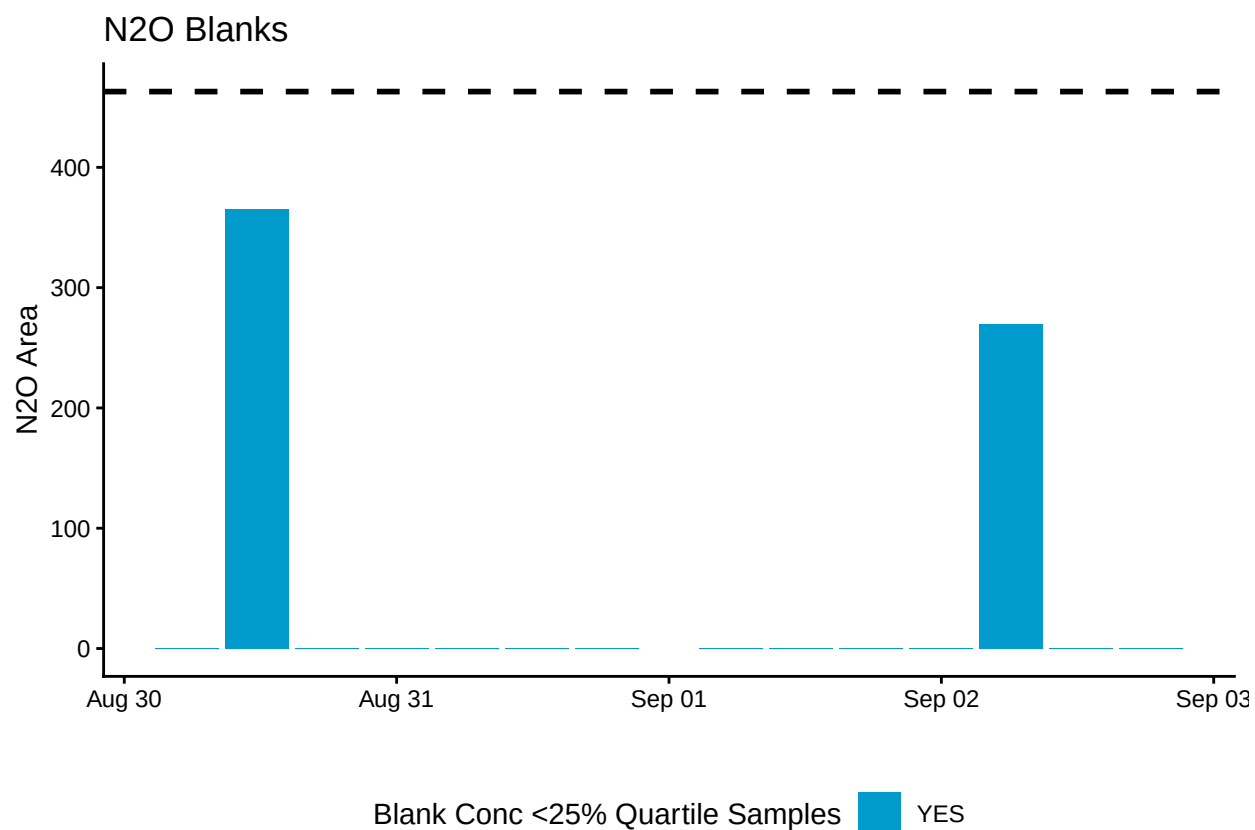
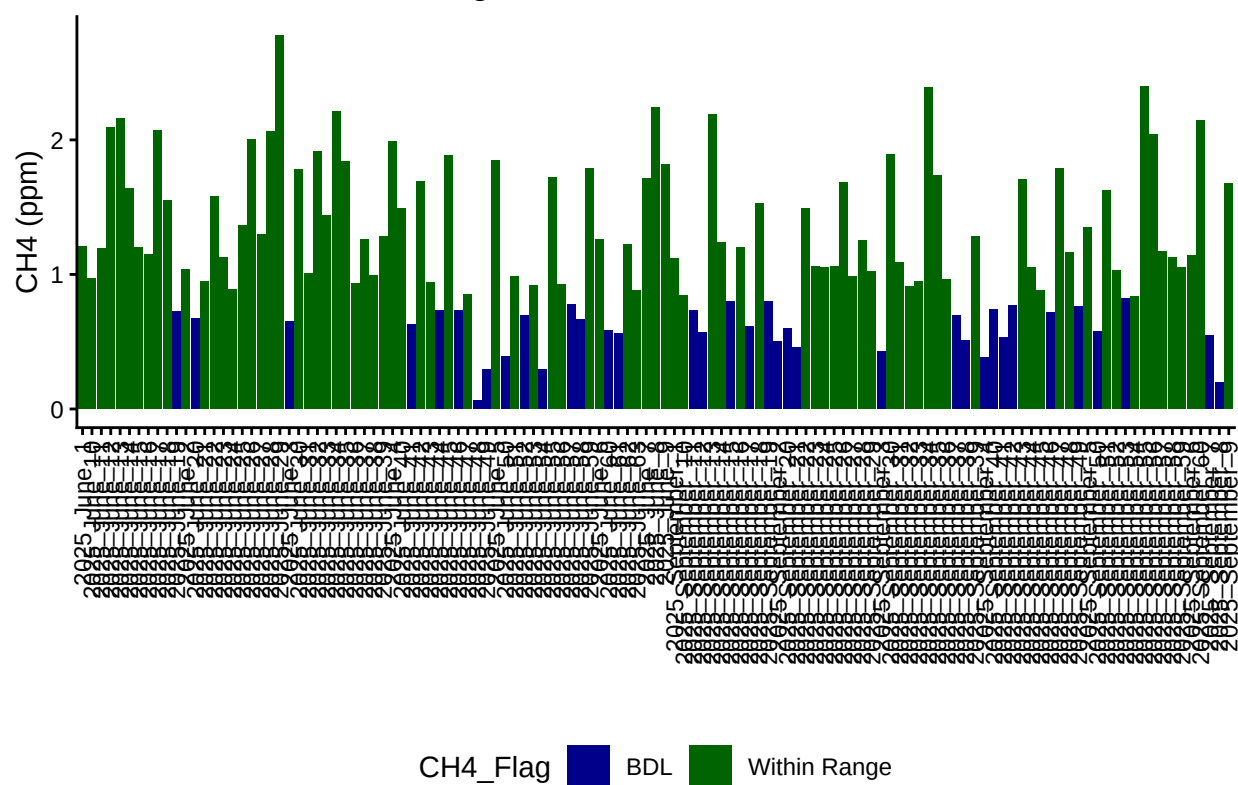


Table 3: Mean Area of Blanks

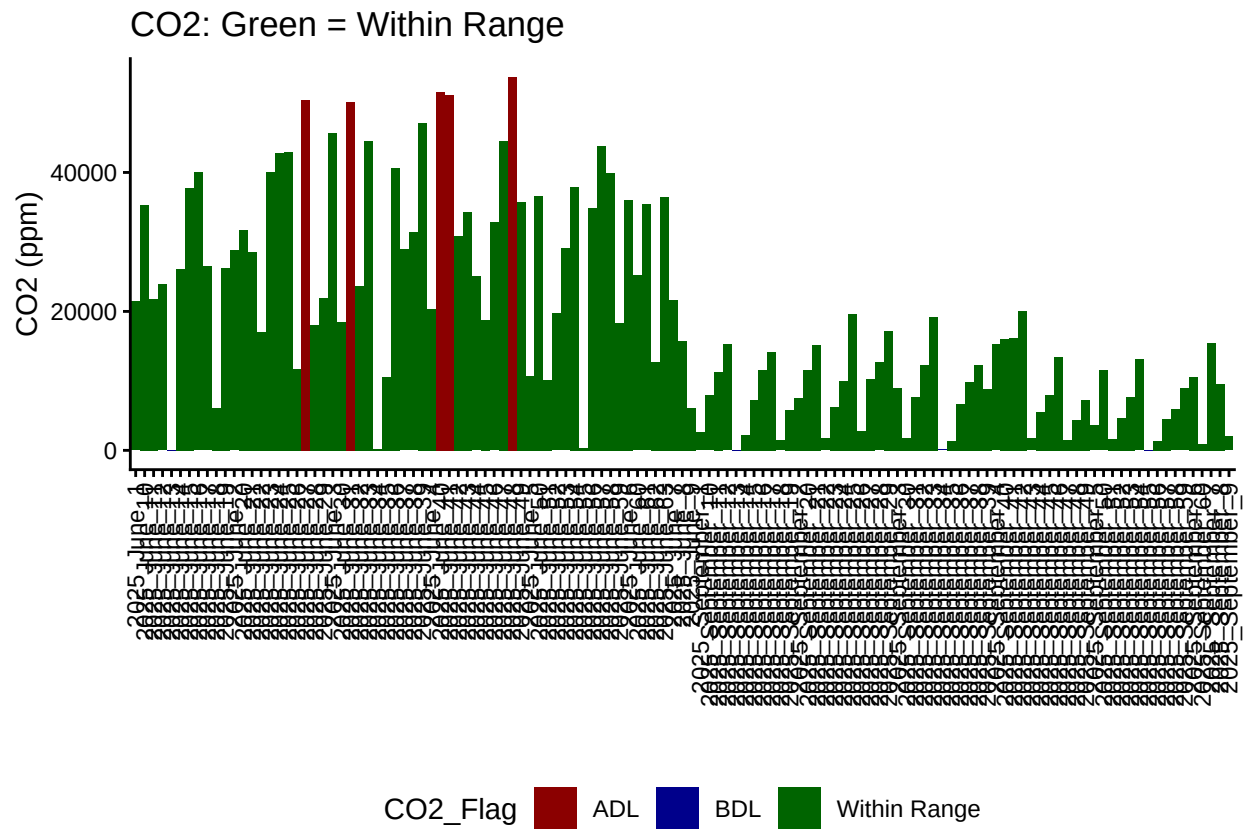
Test	Blank_Mean_Area
N2O	45.3571

10 Check Range - CH4

CH4: Green = Within Range

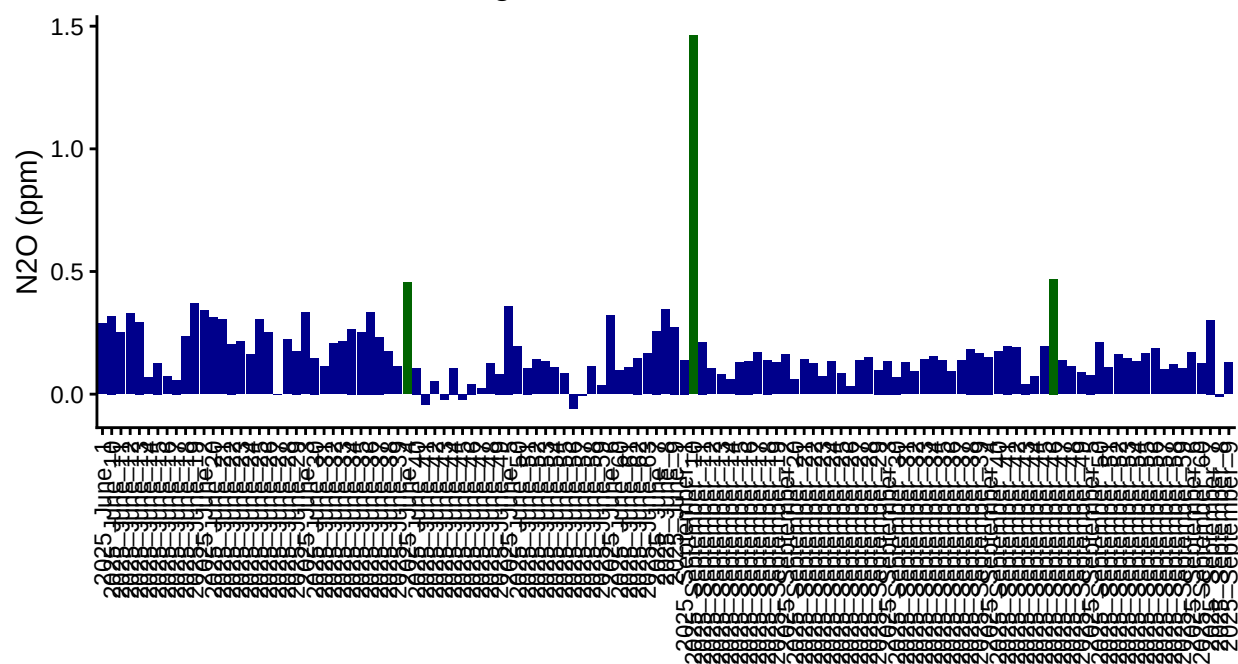


11 Check Range - CO2



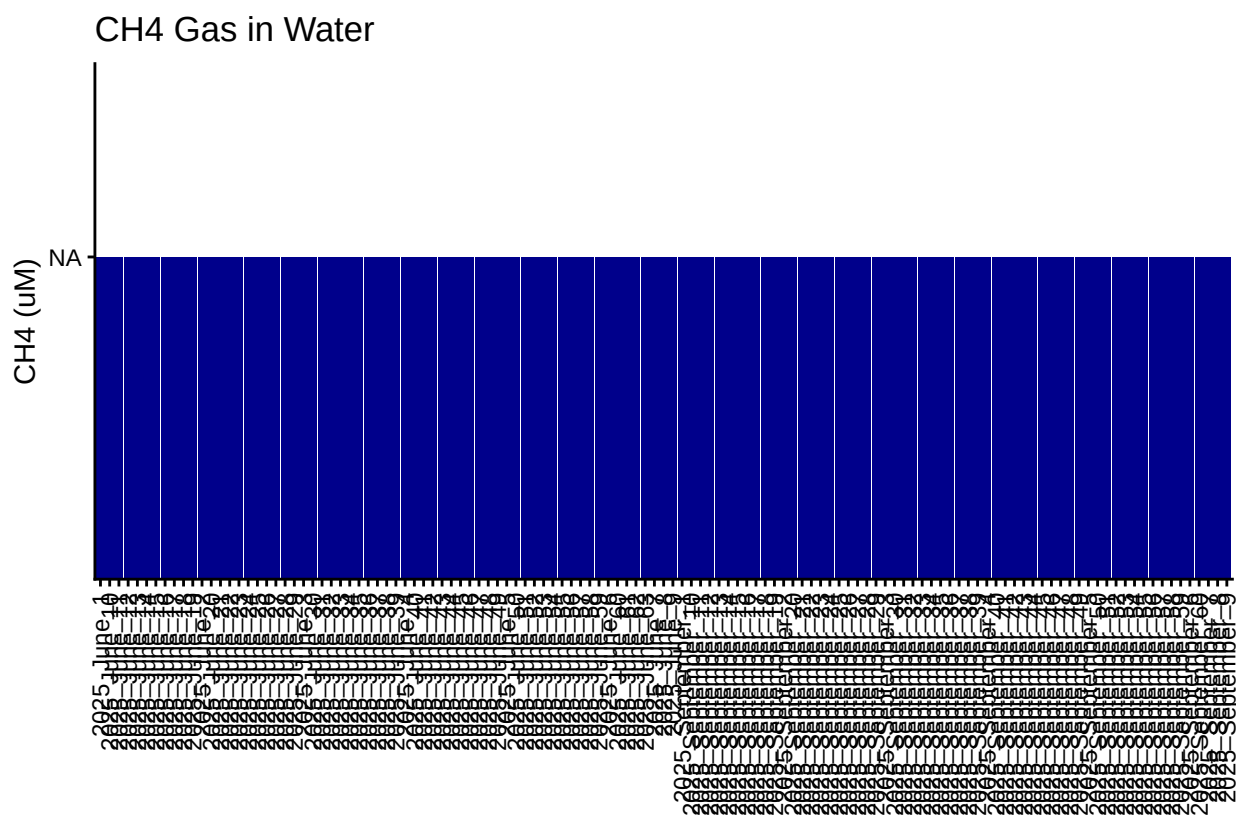
12 Check Range - N2O

N2O: Green = Within Range

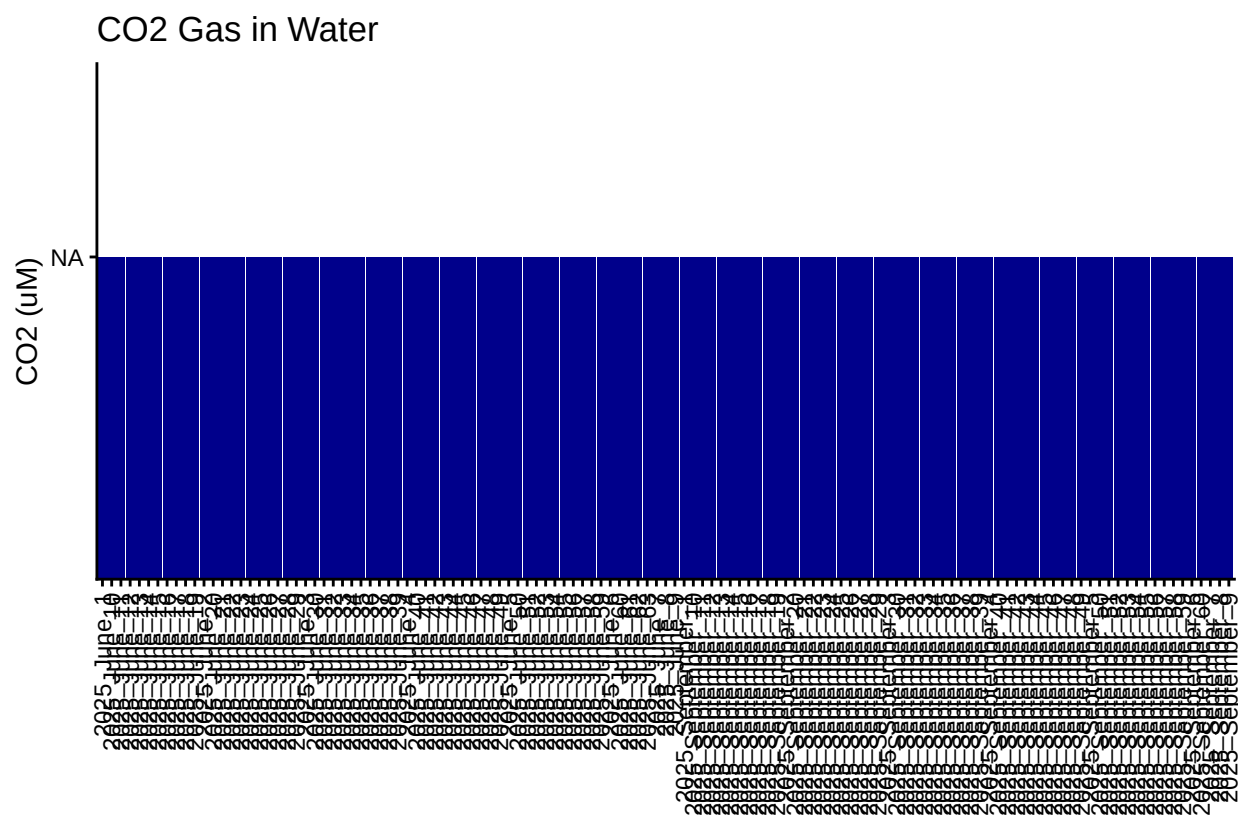


N2O_Flag BDL Within Range

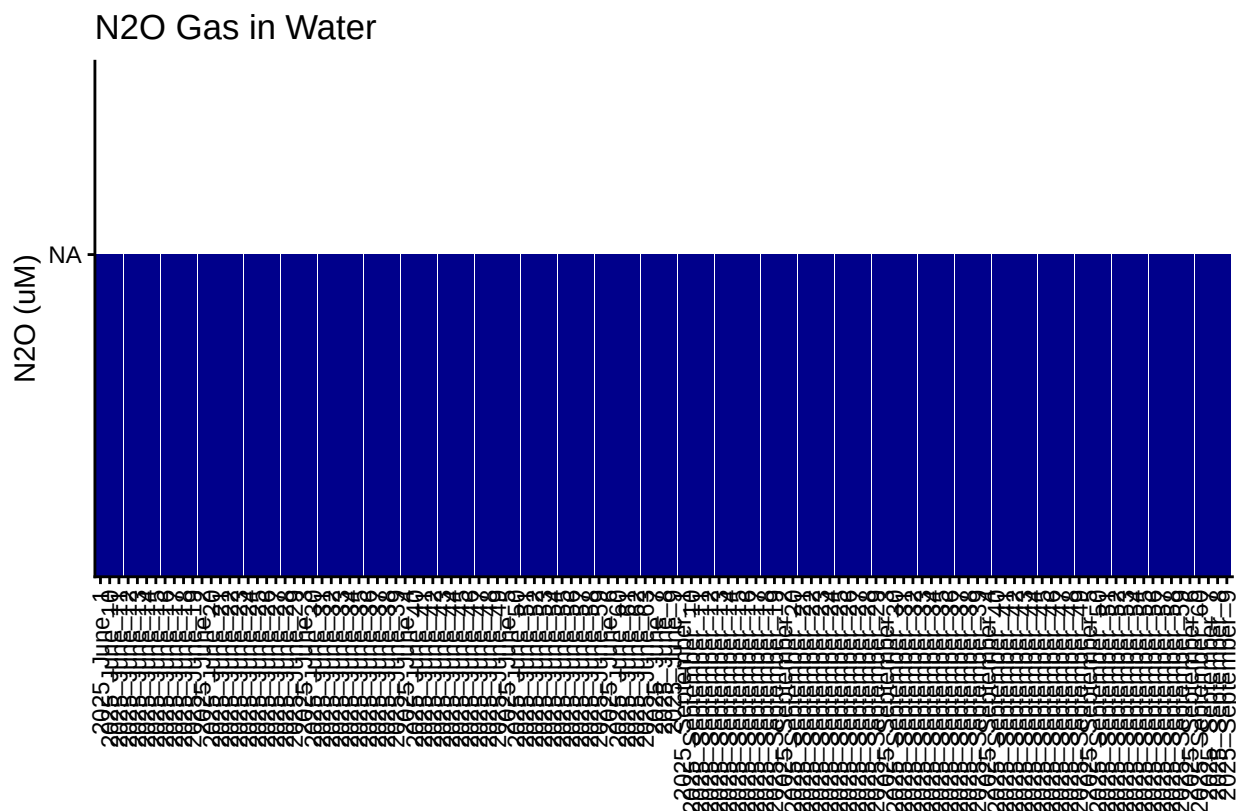
13 Calculate gas in water - CH₄



14 Calculate gas in water - CO₂



15 Calculate gas in water - N2O



16 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

16.1 Organize Data

```
##      Sample_ID  Type   Run_Date CH4_Area Conc_CH4      CH4_RTime_Flag CO2_Area
## 1 2025_June_1 Sample 2026-08-31    364      NA Correct - CH4 RTime 34570
## 2 2025_June_2 Sample 2026-08-31    315      NA Correct - CH4 RTime 46210
## 3 2025_June_3 Sample 2026-08-31    203      NA Correct - CH4 RTime 72803
## 4 2025_June_4 Sample 2026-08-31    587      NA Correct - CH4 RTime 32885
## 5 2025_June_5 Sample 2026-08-31    546      NA Correct - CH4 RTime 17751
## 6 2025_June_6 Sample 2026-08-31    378      NA Correct - CH4 RTime 57608
##      Conc_CO2      CO2_RTime_Flag N2O_Area Conc_N2O      N2O_RTime_Flag
## 1      NA Correct - CO2 RTime    567      NA Correct - N2O RTime
## 2      NA Correct - CO2 RTime    647      NA Correct - N2O RTime
## 3      NA Correct - CO2 RTime    632      NA Correct - N2O RTime
## 4      NA Correct - CO2 RTime    816      NA Correct - N2O RTime
## 5      NA Correct - CO2 RTime    673      NA Correct - N2O RTime
```

```

## 6      NA Correct - CO2 RTime      615      NA Correct - N2O RTime
##  FID_Peak_Flag ECD_Peak_Flag CH4_Conc_ppm      CH4_Flag CO2_Conc_ppm
## 1      Correct      Correct      1.2093868 Within Range      21409.20
## 2      Correct      Correct      1.0382828 Within Range      28788.29
## 3      Correct      Correct      0.6471879      BDL      45646.74
## 4      Correct      Correct      1.9880847 Within Range      20341.00
## 5      Correct      Correct      1.8449160 Within Range      10746.91
## 6      Correct      Correct      1.2582737 Within Range      36013.98
##      CO2_Flag N2O_Conc_ppm      N2O_Flag CH4_Conc_uM CO2_Conc_uM N2O_Conc_uM
## 1 Within Range      0.2875559      BDL      NA      NA      NA
## 2 Within Range      0.3418236      BDL      NA      NA      NA
## 3 Within Range      0.3316484      BDL      NA      NA      NA
## 4 Within Range      0.4564641 Within Range      NA      NA      NA
## 5 Within Range      0.3594606      BDL      NA      NA      NA
## 6 Within Range      0.3201165      BDL      NA      NA      NA

##  Sample1 Sample_Year Sample_Month Sample_Day Sample_Time Time_Zone G_W
## 1      SGW      2025      June      20      1118      EDT      G
## 2      SGW      2025      June      20      1119      EDT      G
## 3      SGW      2025      June      20      1121      EDT      G
## 4      SGW      2025      June      20      1122      EDT      W
## 5      SGW      2025      June      20      1110      EDT      G
## 6      SGW      2025      June      20      1111      EDT      G
##  Sample_Plot Sample_Grid Sample_Depth      Field_Notes Lab_Notes CH4_Conc_ppm
## 1      Control      C3      10      NA      NA      1.2093868
## 2      Control      C3      20      NA      NA      1.0382828
## 3      Control      C3      35      NA      NA      0.6471879
## 4      Control      C3      50 60 mL atm refill      NA      1.9880847
## 5      Control      C6      10      NA      NA      1.8449160
## 6      Control      C6      20      NA      NA      1.2582737
##  CH4_Conc_uM CH4_Conc_Flag CO2_Conc_ppm CO2_Conc_uM CO2_Conc_Flag N2O_Conc_ppm
## 1      NA      Within Range      21409.20      NA      Within Range      0.2875559
## 2      NA      Within Range      28788.29      NA      Within Range      0.3418236
## 3      NA      BDL      45646.74      NA      Within Range      0.3316484
## 4      NA      Within Range      20341.00      NA      Within Range      0.4564641
## 5      NA      Within Range      10746.91      NA      Within Range      0.3594606
## 6      NA      Within Range      36013.98      NA      Within Range      0.3201165
##  N2O_Conc_uM N2O_Conc_Flag      QAQC_Flags
## 1      NA      BDL N2O_R2 is Bad; N2O Check Standards out Of Range
## 2      NA      BDL N2O_R2 is Bad; N2O Check Standards out Of Range
## 3      NA      BDL N2O_R2 is Bad; N2O Check Standards out Of Range
## 4      NA      Within Range N2O_R2 is Bad; N2O Check Standards out Of Range
## 5      NA      BDL N2O_R2 is Bad; N2O Check Standards out Of Range
## 6      NA      BDL N2O_R2 is Bad; N2O Check Standards out Of Range
##  Analysis_rundate
## 1      2026-08-31
## 2      2026-08-31
## 3      2026-08-31
## 4      2026-08-31
## 5      2026-08-31
## 6      2026-08-31
##
##      Run_notes
## 1      Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected
## 2      Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected

```

```
## 3 Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected
## 4 Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected
## 5 Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected
## 6 Run 8/31 - Blank8 -> CheckStd, Run 9/02 - Blan1,2,4,5,11 -> ChkStd -- Corrected
```

16.2 Visualize Data CH4

```
## Visualize Data CH4
```

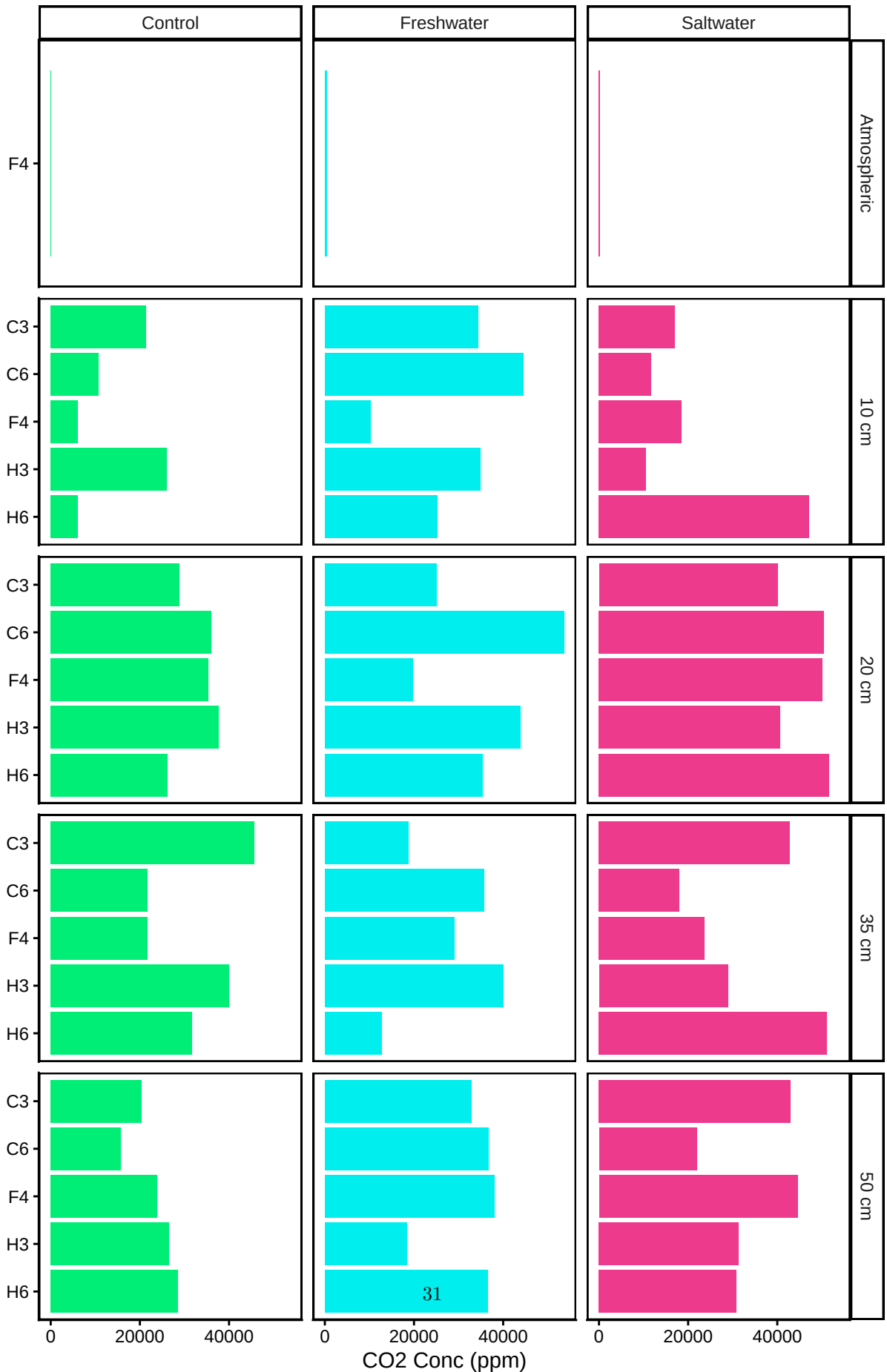
TEMPEST: Soil Gas Well CH4



16.3 Visualize Data CO2

```
## Visualize Data CO2
```

TEMPEST: Soil Gas Well CO2



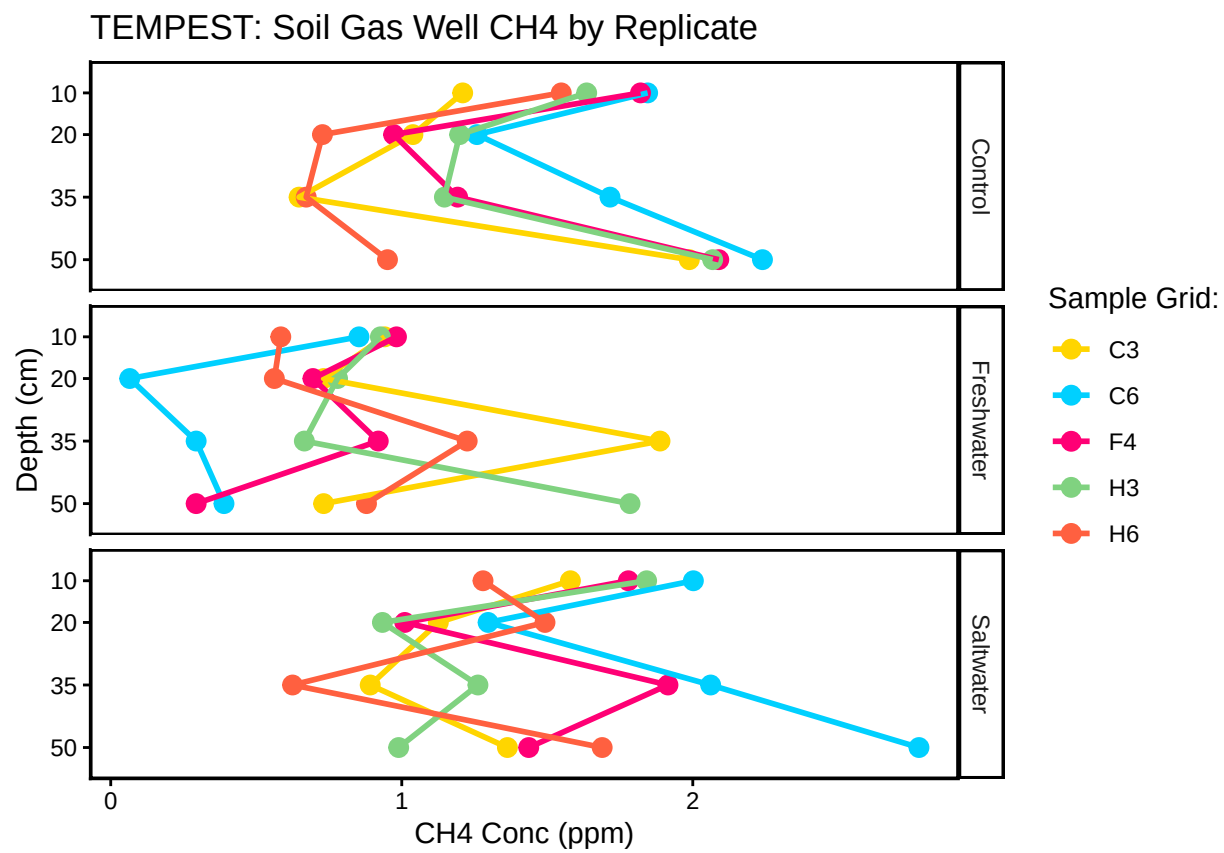
16.4 Visualize Data N2O

```
## Visualize Data N2O
```

TEMPEST: Soil Gas Well N2O

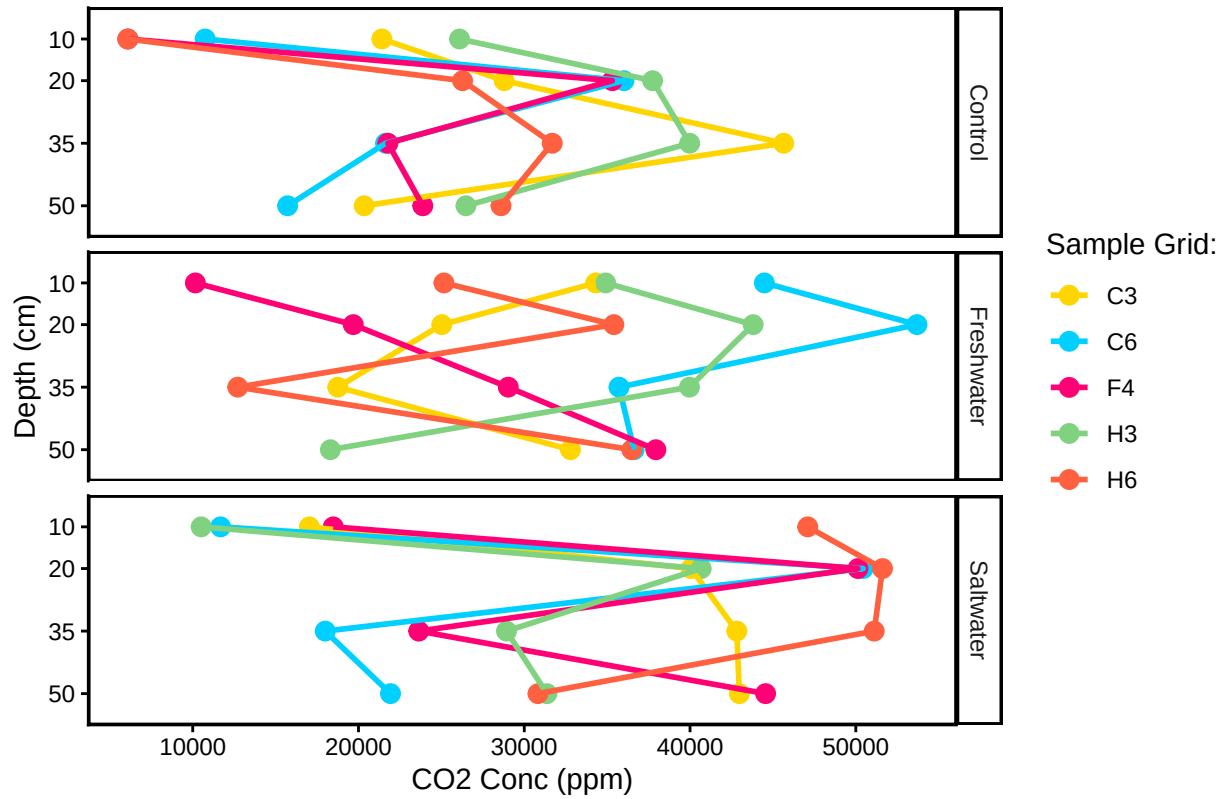


#Visualize Data CH4 by Replicate



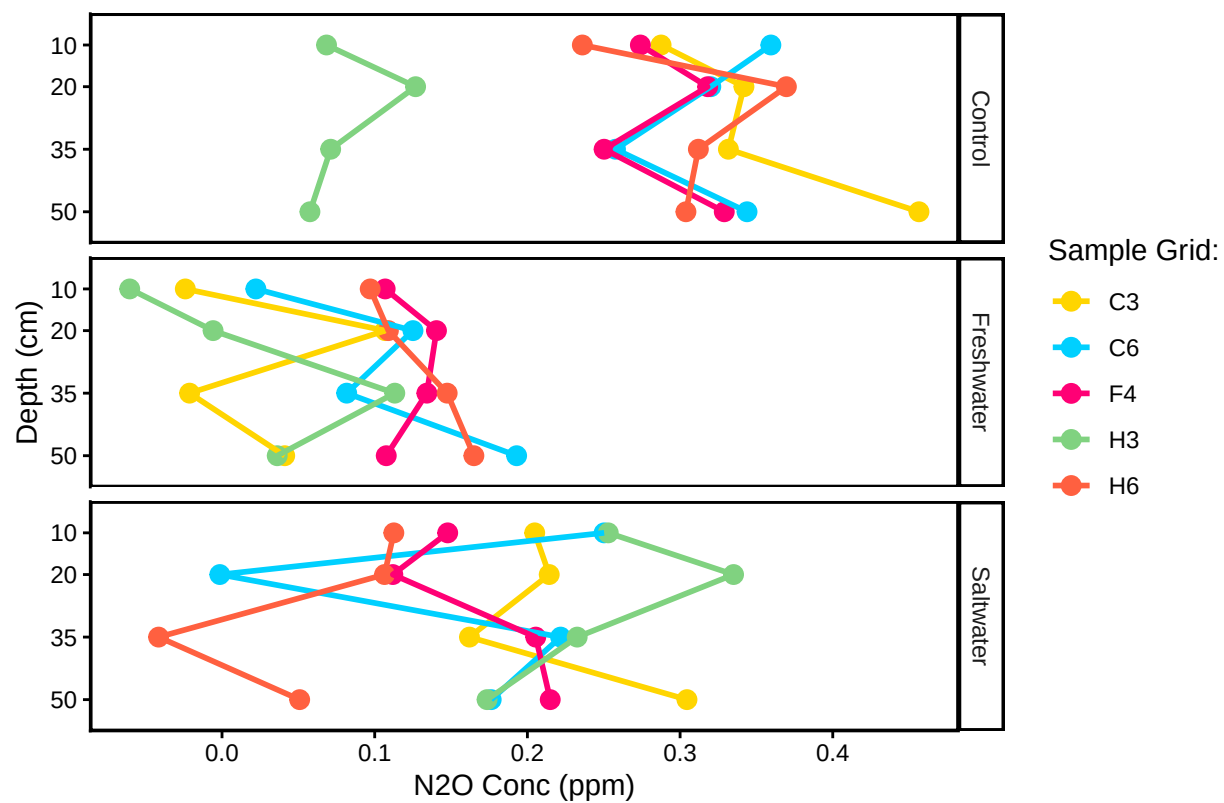
#Visualize Data CO2 by Replicate

TEMPEST: Soil Gas Well CO2 by Replicate



#Visualize Data N2O by Replicate

TEMPEST: Soil Gas Well N2O by Replicate



##Export Data

#end